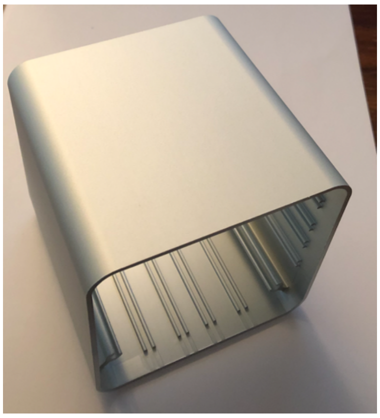
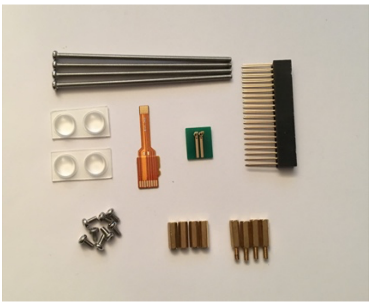
PIXIS Assembly Instructions

Raspberry Pi 3A+, Pixis CB-1Carrier Board and Pixis 90EXT Case.

1. Start with these parts. Use the Raspberry Pi Imager software to prepare a microSD card with a bootable OS for step 8. Ensure you enable ssh and enter your home network Wifi SSID and password as you will need these later.   https://raspberrypi.com/software/


CB-1


90EXTS (or EXTB)

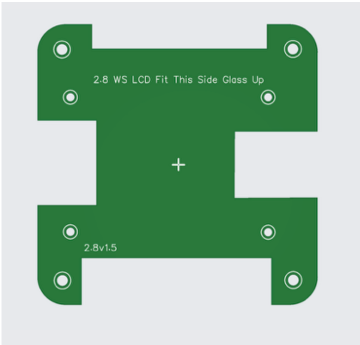

SSK1


PI3A+


BP3

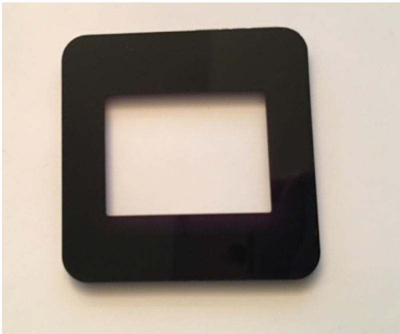

BP (If No LCD)

OPTIONAL LCD PARTS

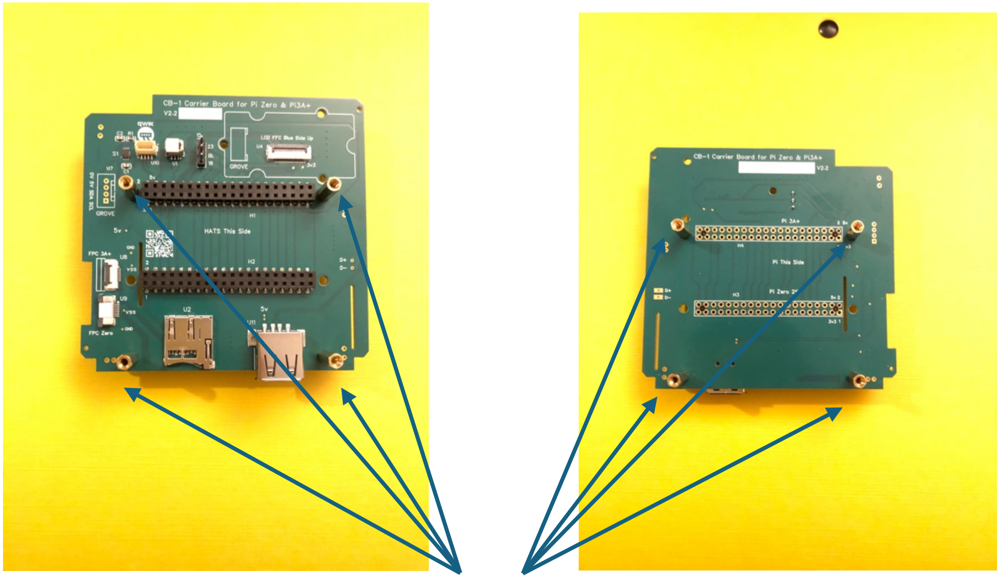


CB-3


2.8WSL or 2.8WSS

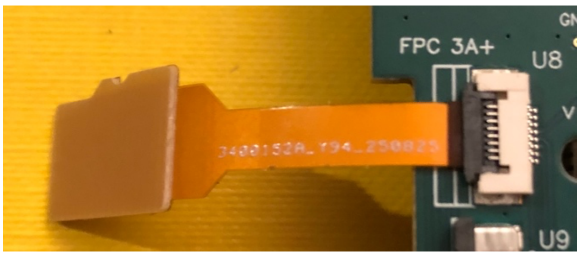

FP28

2. Fix the 8 x 12 mm spacers from the CB-1 Kit to the CB-1 board.
   a) 4 x Top HAT side spacers are male – female.
   b) 4 x Bottom Pi side spacers are female – female.

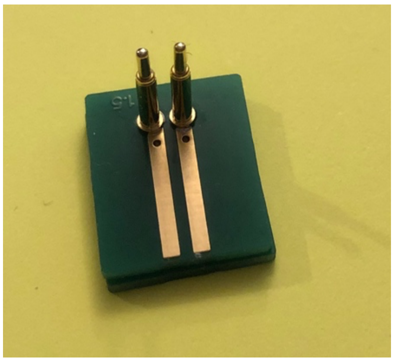


8 x 12 mm spacers

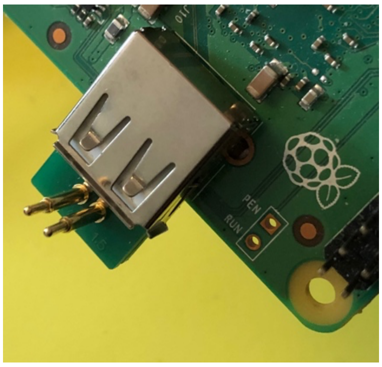
3. Fit the microSD Card flex cable to the CB-1 connector U8.
   a) Flip up the black lever of FPC connector U8.
   b) Insert the flexi cable with gold fingers facing down.
   c) Ensure black FPC stiffener aligns with the legend inner white line.
   d) Close the black lever and press firmly into place.



4. Fit the USB Bridge board to the Pi 3A+ USB socket. Pogo pins point upwards.

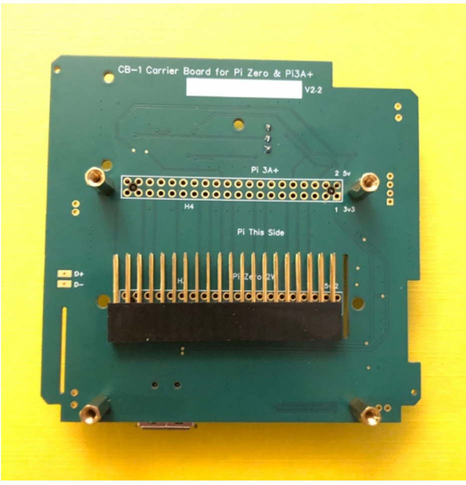


USB Bridge Board

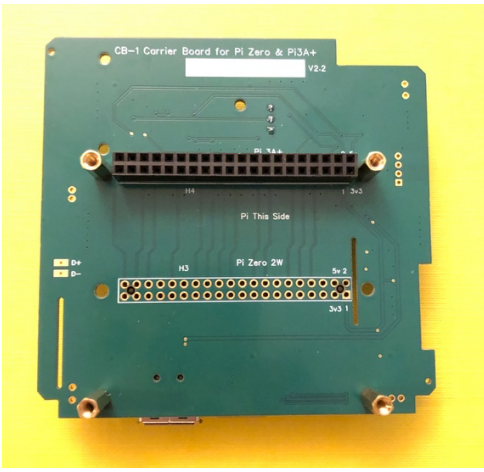


Plugged into Pi 3A+ USB

5. Fit the 40-pin header to the CB-1 board pushing carefully the long pins into the connector holes marked H4 and Pi 3A+.



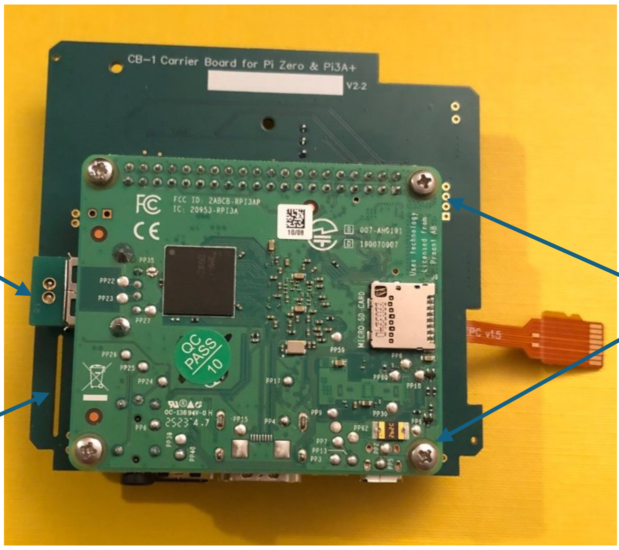
Pi 3A+ with 40-pin Header



40-pin Header inserted

NB **If you are installing a DSI LCD then now is a good time to insert the DSI cable into the Pi 3A+ DSI FPC connector prior to step 6.**

6. Fit the Pi 3A+ to CB-1 with 4 x M2.5 screws. Note the CB-1 board has the legend “Pi This Side”. If fitting a DSI LCD, pass the cable through the cable slot on CB-1. Ensure the USB Bridge pogo pins line up with their landing pads on CB-1 before fully tightening the screws.

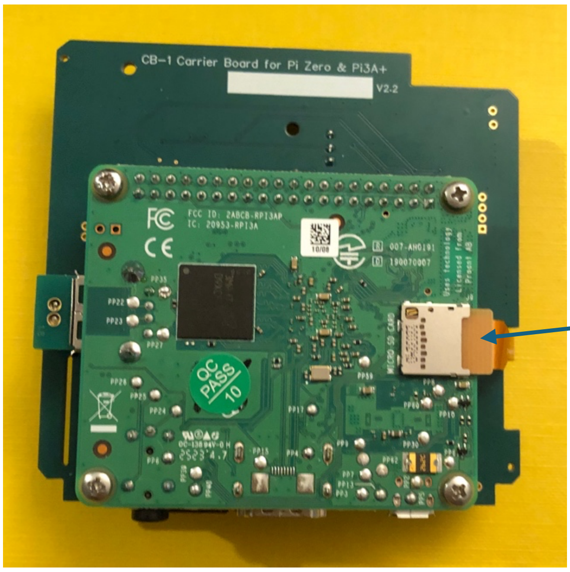


Line up pogo pins

DSI FPC cable slot

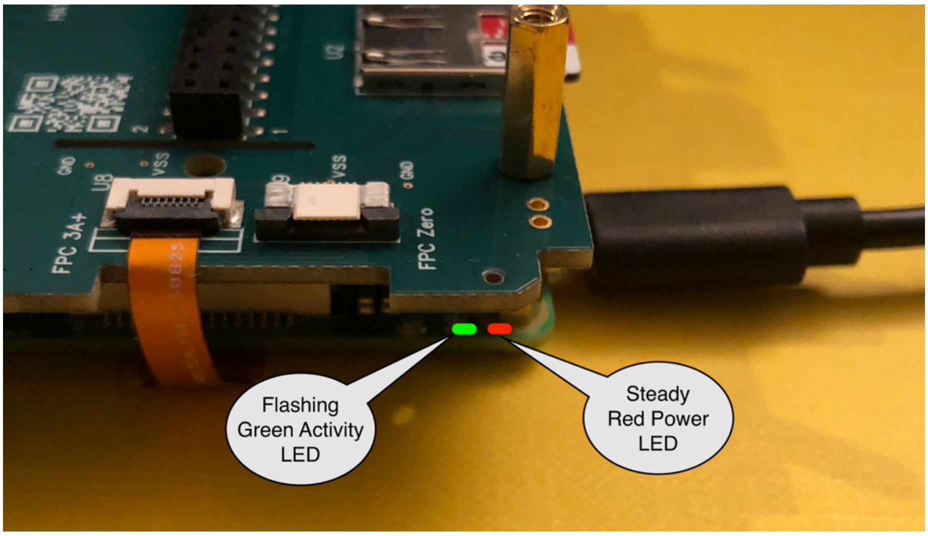
4 x M2.5 Fixing Screws

7. Fold over and fit the microSD Card flex cable into the Pi 3A+ microSD Card holder. The cable length is intentionally tight to ensure the flexi does not foul the inside of the case when the assembly is slid into the case. This is a little fiddly so take care and take your time to ensure a good fit. Refer to step 3 images for the correct flexi cable orientation.



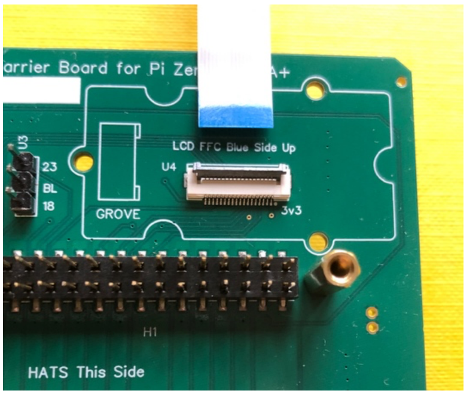
Insert SD Card stiffener

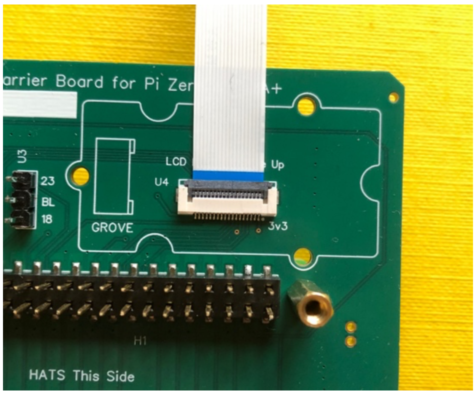
8. To ensure the cable has been inserted correctly, now is a good time to fit a known-good, OS-loaded microSD card to the CB-1 board and power up the assembly to ensure the Pi 3A+ boots OK. A flashing green activity LED on the Pi 3A+ in the first 30 seconds is a good sign. If no flashing green activity LED, then power down your Pi and double check the flexicable connections.



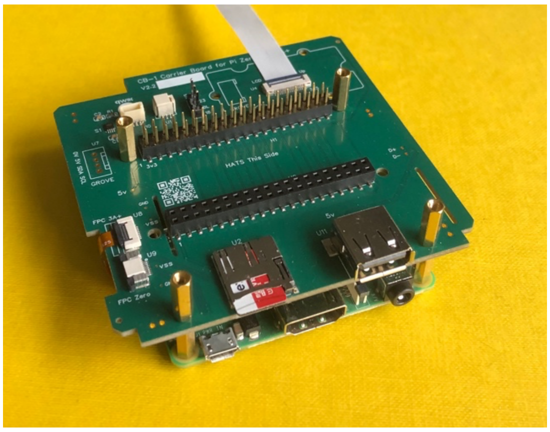
9. If you are installing a 2.8-inch SPI LCD please attach the SPI LCD flat-flex (FFC) cable to U4 on CB-1. Flip the black catch up, insert the FPC ensuring the blue stiffener is facing up and then close the catch down firmly. Check the jumper on U3 is BL-18. This sets the LCD backlight to the default GPIO 18. Should you need to use GPIO 18 in your project for something else, then fit the jumper at BL-23. This alters the backlight enable pin to GPIO 23. Remember to edit the config.txt file accordingly.

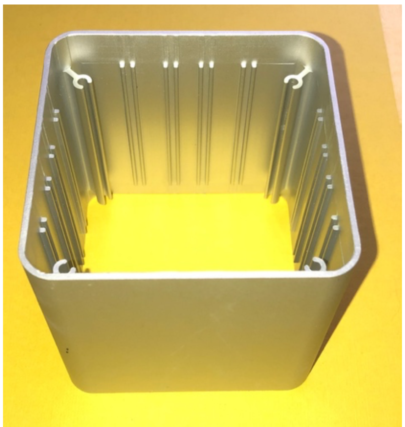
See https://github.com/PIXISREPO/PIXIS-CODE/ for the display driver and the exact Landscape/Portrait config.txt settings.





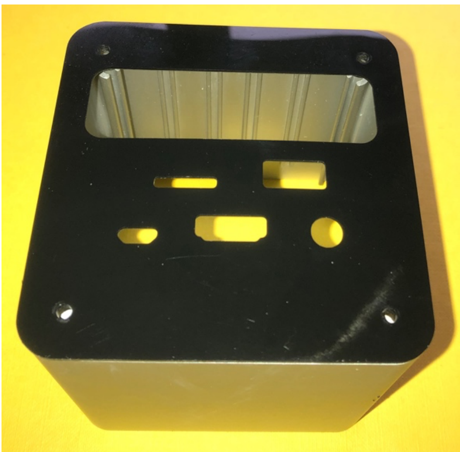
10. Check that your CB-1assembly should now look like this and is ready to slide into the case. As the case is a square profile and the PCB sliders are all equal on each side it does not matter which way round you place it. What does matter is which way up the case is. The PCB sliders are machined down into the body of the case with one end machined deeper than the other. The shallow end is where you will mount the front panel with the optional LCD and the deep end is where you will mount the back panel for the input and output connectors.





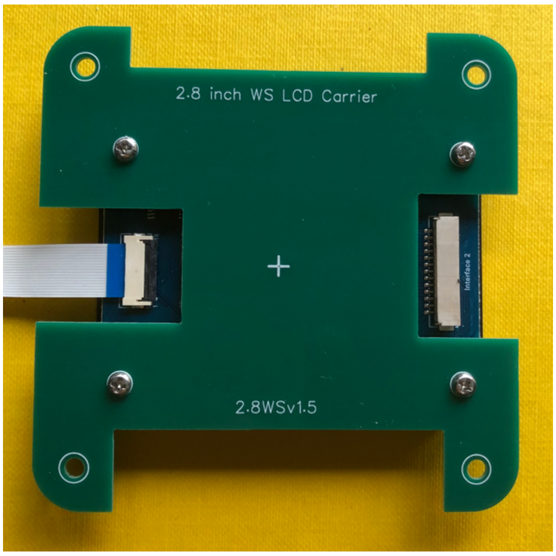
11. Fit the back panel to the deep end of the case (A), Make sure it is pushed in securely then turn the case over so the top is open upwards (B). Then slide the CB-1 assembly into the case using the second from the top set of runners.

(A)

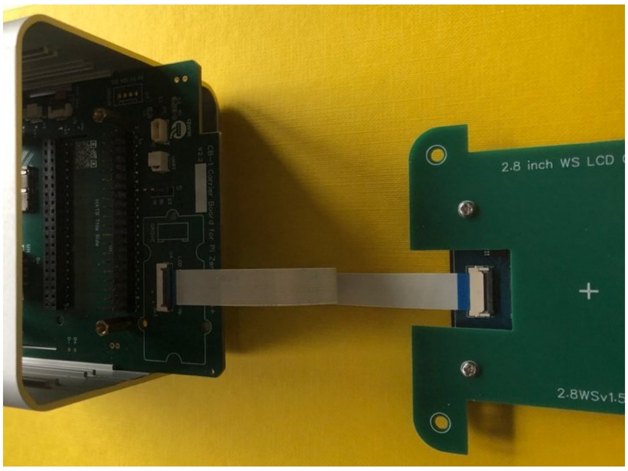


(B)

12. If you are fitting a 2.8-inch SPI LCD now is the time to assemble the LCD to the carrier board using the four black M2.5 flat head screws. Ensure the LCD is mounted on the correct side of the carrier board 'Glass side up'. Turn the LCD and carrier board over and fit the FFC to the LCD FFC connector. Ensure the blue side of the FFC is facing upwards and is visible after fitting. Flip the back side of the LCD FFC connector lever up – insert the blue end of the FFC carefully and then close the lever down firmly. The finished LCD sub-assembly should now look like this.

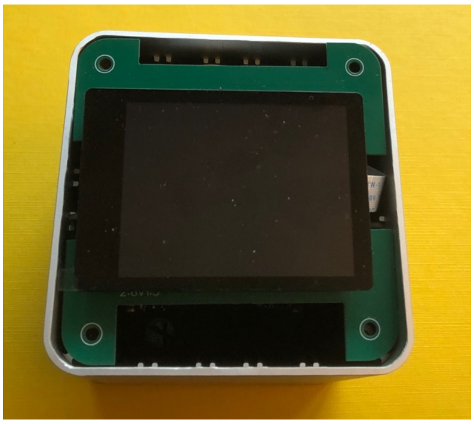


13. Attach the other end of the LCD FFC to the FFC connector U4 on CB-1. You can slide the carrier board out of the case slightly to make that connection easier. The blue side of the FFC should be visible at both ends after fitting.



14. Place the LCD sub-assembly into the case top. The silk screen legends on the LCD carrier board should be right way up i.e. not upside down when the case is held upright and the Pi 3A+ connectors are at the bottom of the case. The LCB Carrier Board will wobble slightly on the Carrier Board pcb edge – do not be concerned, when the front panel is fitted to the case it will position the LCD correctly

The front and back view of the assembled case should now look like this.



15. Fit the front panel onto the case over the LCD. There are vent slots for air intake on one edge of the LCD panel, make sure these are on the bottom edge of the box.

An elastic band or some masking tape is useful to temporarily hold the front and back panels in place whilst the four long fixing bolts are inserted into the case from the back. Screw each bolt down finger tight only and ensure the front and back panels are aligned with the case and that the 1 mm lips on the inside edge of the panels are inside the case.

When screwing down the four bolts take care not to overtighten.

The finished assembly should now look like this.

The HAT blanking panel is NOT shown fitted.

16. When the assembly is complete, power up the CB-1 and install the Drivers for the Waveshare 2.8-inch SPI LCD - SKU 27579. The PIXIS display drivers and the required config.txt settings for both Landscape and Portrait orientation are available from the PIXIS-CODE GitHub repository.

    https://github.com/PIXISREPO/PIXIS-CODE/

    Follow the README instructions, download the raw binaries and copy them to the /usr/lib/firmware directory. Add the corresponding Landscape or Portrait configuration text block to the [all] section of /boot/firmware/config.txt.

    The standard PIXIS configuration uses U3 at BL-18 (backlight-gpio=18)
    If you require GPIO 18 for other purposes then move the U3 jumper
    To BL-23. Remember to change the backlight setting to backlight-gpio=23.

17. Should you wish to 3D print or CNC machine either front ot back panels to accommodate different displays or HAT connector openings then CAD and CAM files for the PIXIS case and panels are available at

    https://github.com/PIXISREPO/PIXIS-CAD/

18. A selection of ready made panel parts are available at the Pixis Store

    https://pixis.uk/products/